

Can a model tell an officer whom to search?

Scoring consent searches in Nashville: three models, four tests, one recommendation

How we evaluate each model

	Performance	Interpret-ability	Stability	Fairness
White box				
XGBoost				
TabPFN				

Then: the trade-offs, and which model to deploy under trustworthy-AI requirements.

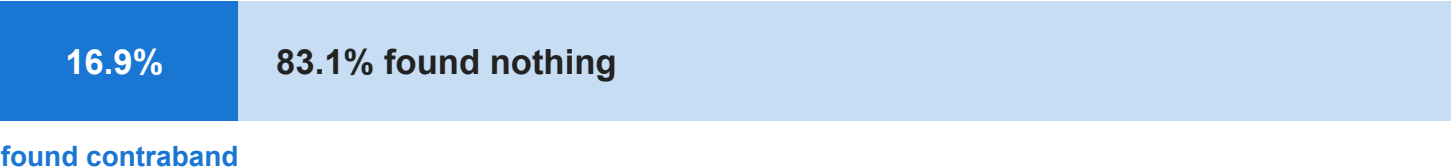
The client's question

Consent searches are MNPD's most discretionary searches, and 83% of them find nothing

Nashville traffic stops to consent searches, 2010–2018



Outcome of the 58,865 consent searches



Drivers searched with consent, by race



The client's question

"Can a scoring model tell our officers which stops are worth a consent request?"

No warrant, no probable cause: the driver agrees. Each fruitless search costs officer time, the driver's time and dignity, and community trust.

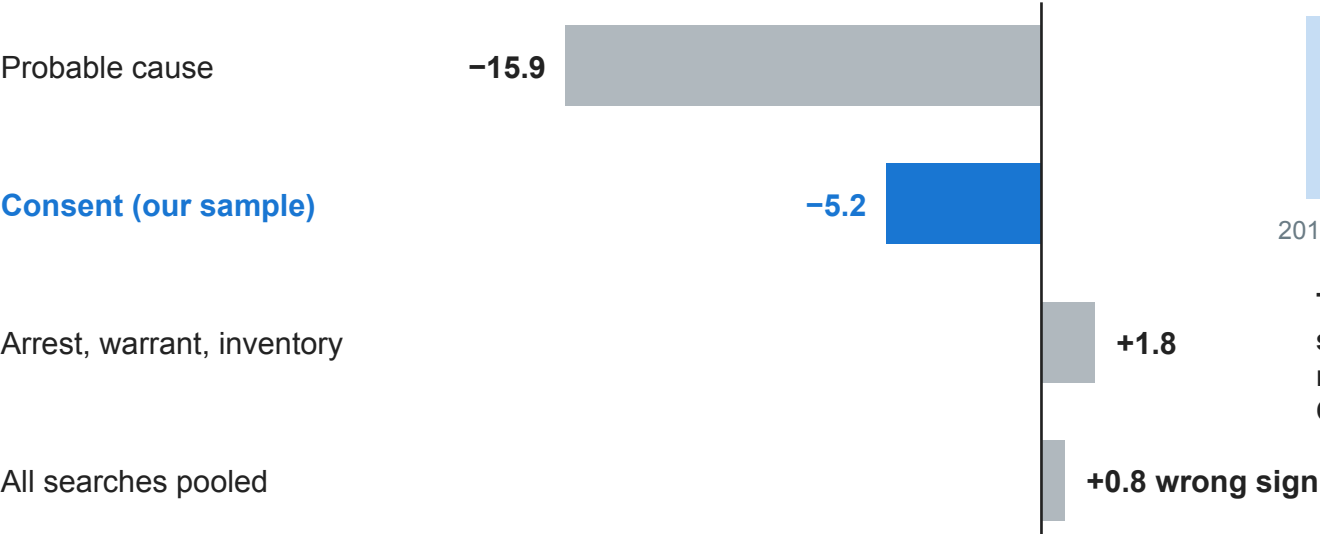
Success criteria (ours to state, theirs to accept)

- 1 Fewer fruitless searches
- 2 Finds preserved
- 3 No discrimination
- 4 Explainable to a driver, a court and the oversight board

Consent search = consent recorded and no arrest, warrant, inventory or plain-view authority. Race as perceived by the officer. MNPD is a role-play client.

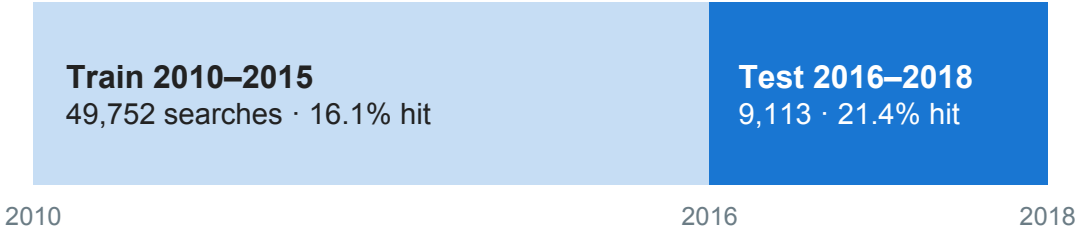
Pooling all searches would reverse the disparity, so we score consent searches only

Hit-rate gap, Black minus white drivers, by legal basis (pp)



Below zero: consent searches of Black drivers find contraband less often, consistent with a lower bar for searching them. Discretionary searches show it, mechanical ones do not, and pooling hides it. The same pattern appears in Illinois (-7.3 pp) and North Carolina (-6.3 pp).

How we test the models



Temporal split: no learning from the future, as in deployment. Consent searches fell from ~9,000 a year (2011–2014) to 2,167 in 2018 as the hit rate rose.
Officer holdout: 20% of officers never seen in training.

Selective labels shape everything

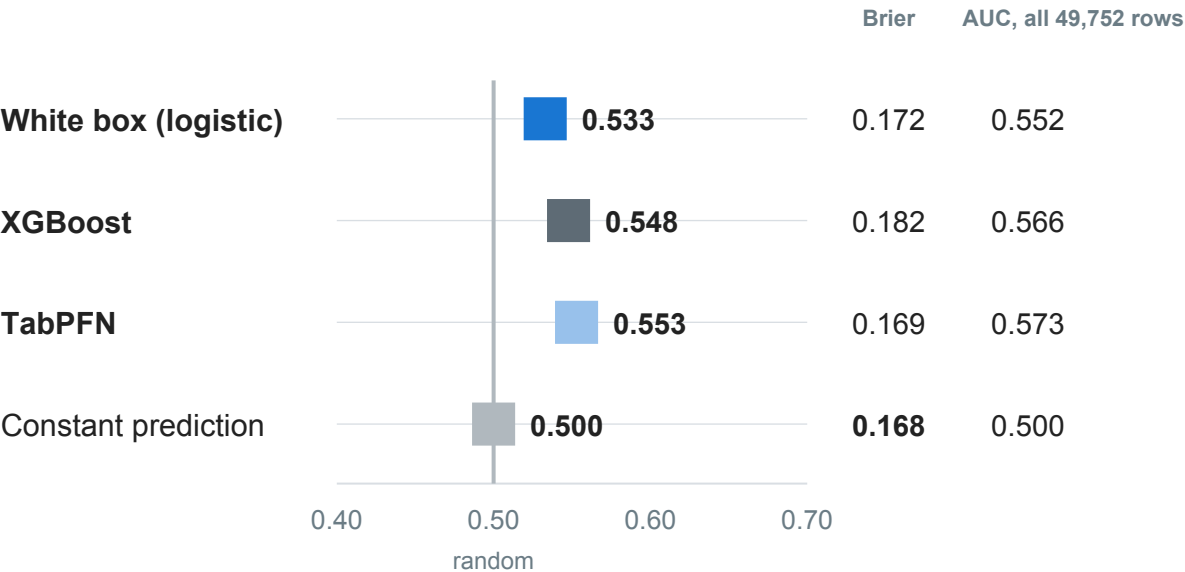
Contraband is observed only when an officer chose to search. Every model learns from officers' decisions, and every number is conditional on a search having been made.

Legal-basis table from the earlier cleaning (58,836 consent searches); our sample is 58,865. Excluding plain-view and arrest cases, the consent gap is -5.16 pp (-4.60 pp with them).

All three models rank searches barely better than chance, and the best adds 82 finds per 2,000 searches

Test AUC on the 9,113 consent searches of 2016–2018

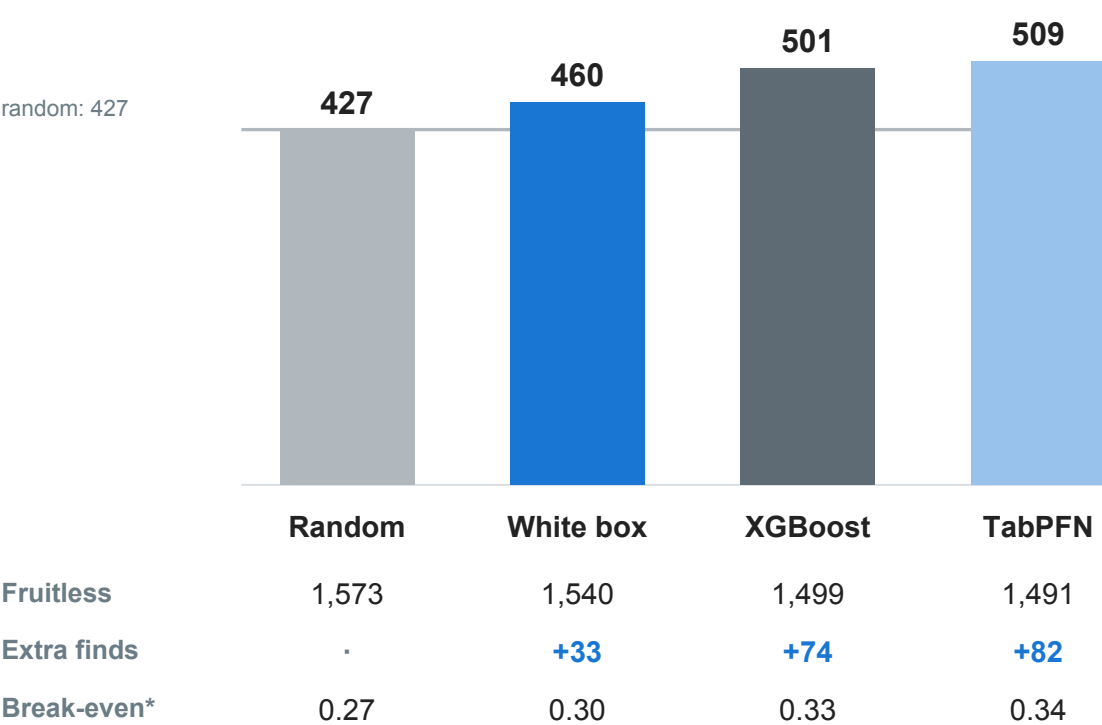
Dots: all three trained on the same random 2,000 of the 49,752 searches of 2010–2015 (the most TabPFN could run on CPU), so only the model differs



DeLong: XGBoost vs TabPFN -0.005 ($p = 0.41$), tied. The white box is below both (-0.015 , $p = 0.03$; -0.020 , $p < 0.001$).
Brier: none beats a constant prediction (0.168). With all 49,752 rows, AUC rises by only 0.02 and the ranking is unchanged.

Finds if MNPD searched only the 2,000 top-scored of the 9,113 test stops

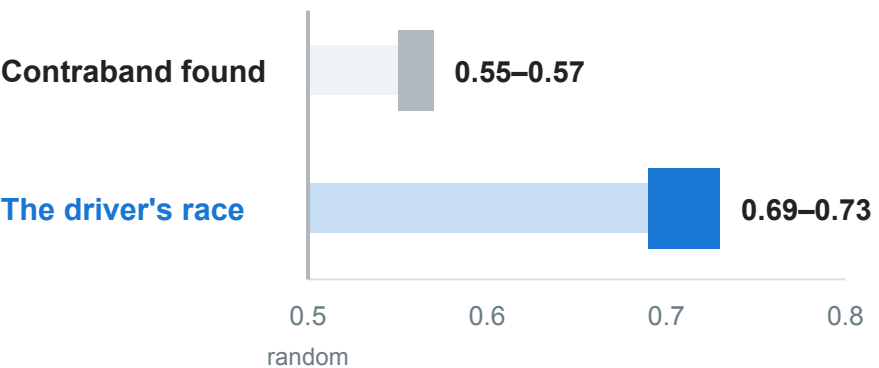
A search budget of $\approx 22\%$; same models as on the left (2,000 training rows)



*Break-even cost ratio = hit rate / (1 - hit rate) of the searches kept: the cost of searching an innocent driver, as a fraction of the value of a find, above which keeping the top 2,000 destroys value. The price is a policy judgement: we show it, we do not pick it.

What little the models learn is mostly the driver's race, and only the white box explains it exactly

What the same stop and driver features predict (test AUC)



73.5% of the white box's above-chance AUC is the driver's race (XPER)

Dropping race does not remove it: the remaining features recover race at AUC 0.69–0.73. The white box's largest coefficient is "driver Hispanic" (odds ratio 0.43, –11 pp).

White box

MAIN DRIVER
Driver Hispanic is the largest coefficient; race is 73.5% of the signal
ONE DECISION EXPLAINED BY
Exact: coefficient × (x – mean)
GLOBAL EXPLANATION
The model is its own explanation

XGBoost

MAIN DRIVER
Flat: precinct, zone, month, age, hour ≈ 0.01 each; race 0.0025
ONE DECISION EXPLAINED BY
Exact TreeSHAP
GLOBAL EXPLANATION
SHAP; a depth-3 surrogate tree reproduces only 39% of the score

TabPFN

MAIN DRIVER
Race is its largest feature (0.038, 15× its weight in XGBoost)
ONE DECISION EXPLAINED BY
Occlusion only: approximate, error cannot be bounded
GLOBAL EXPLANATION
Permutation importance; no native attribution

Same accuracy, contradictory explanations: XGBoost's and TabPFN's importance rankings correlate at –0.32.

XPER: the course's Shapley decomposition of AUC above 0.5, exact over 1,024 coalitions. Importances: permutation, test set. The surrogate tree was fitted on the Act II XGBoost.

Retrain the model on a slightly different sample of the same years, and its shortlist barely survives

93%

of the 200 highest-scored searches change identity when we retrain on a different random sample of the same years

Same driver, same officer, same years of data — just a different random split — and the list of "high-risk" stops is almost a different list.

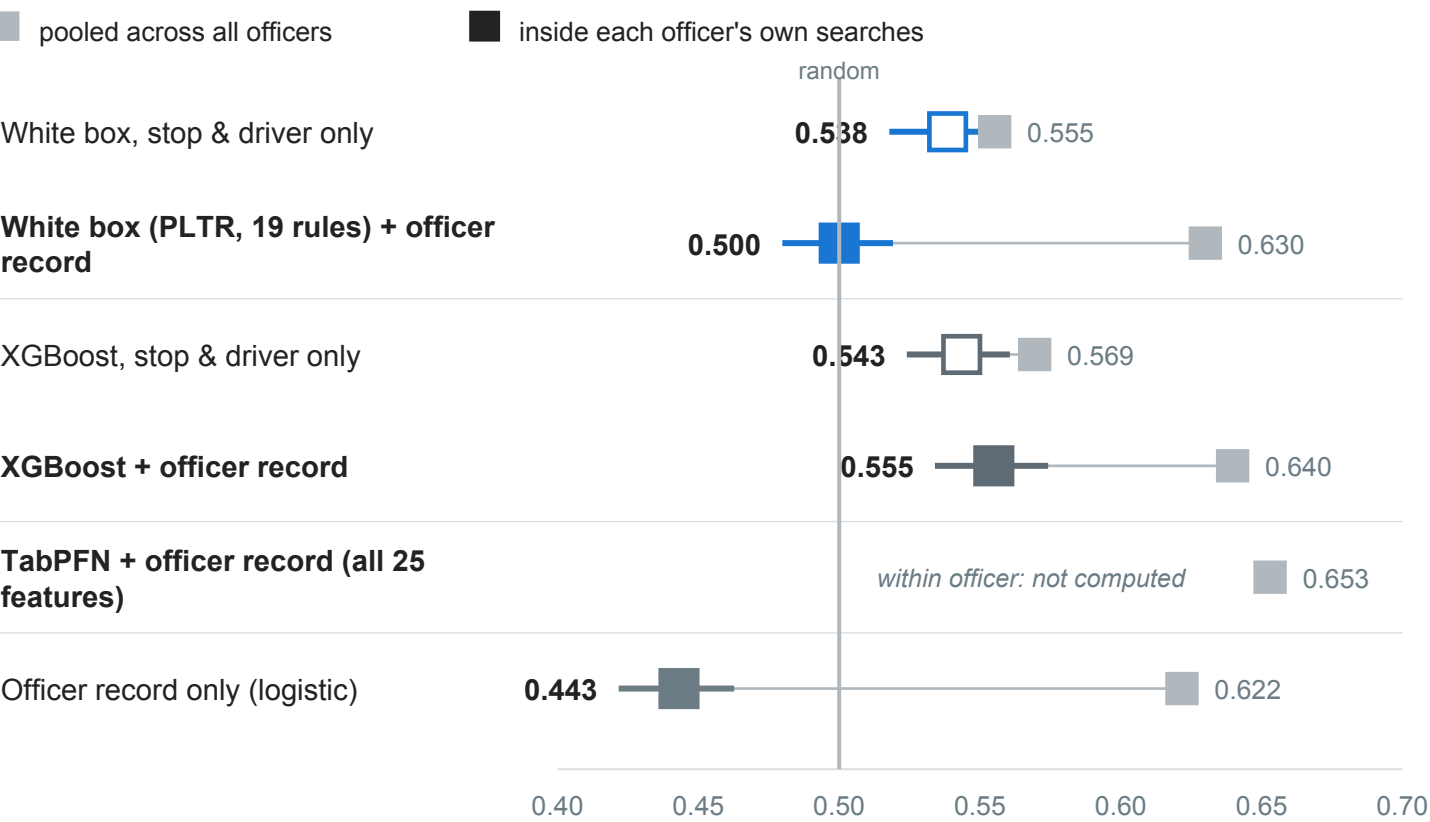
Top-200 overlap is only 2.3× what pure chance would give: barely better than picking names out of a hat.

TabPFN is the most sensitive of the three: which past searches happen to be in its memory moves its own score by more than the entire gap between all three models.

A shortlist that changes every time you retrain it is not something officers, or a court, can rely on.

The officer's past record lifts every model by 0.07, yet inside one officer's own searches the scores fall back to chance

Pooled AUC vs AUC inside each officer's own searches, test 2016–2018 (95% CI)



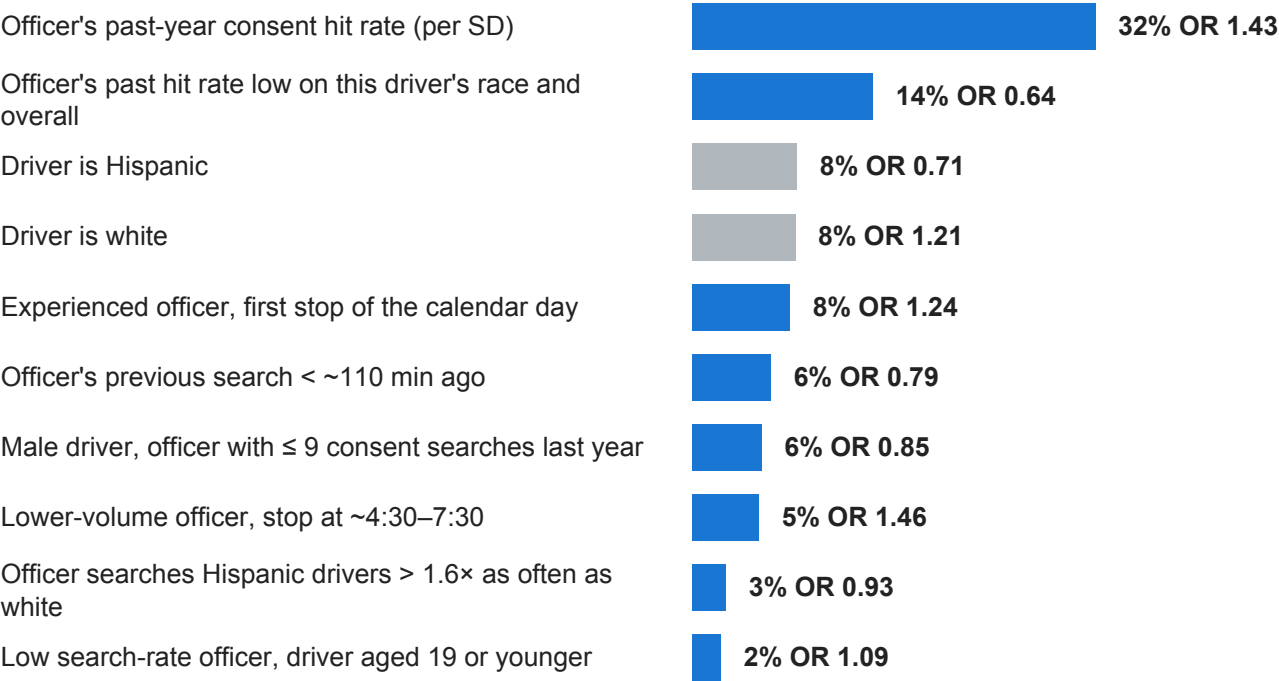
The gain is real
+0.070 [+0.056, +0.084] for XGBoost; DeLong $p = 6.6 \times 10^{-23}$
It holds on officers never seen in training (XGBoost 0.643, white box 0.629), in every test year (+0.05 to +0.09) and over 5 other officer draws

But it lies between officers
With the record, 52%–60% of the score's variance is between officers (24% without)
The record tells you which officer, never which driver
So we use it to explain outcomes, never as an input to a deployed score.

Within-officer AUC: AUC inside each officer's own test searches, 209 officers with ≥ 10 searches and both outcomes (7,005 searches), weighted by searches; CI resamples officers. The record uses only events strictly before the stop (verified on 400 stops).

The white box reads like a profile of the searcher: 81% of its weight is about the officer

The white box's 10 heaviest rules: share of importance (odds ratio)



■ involves the officer ■ driver or stop only

XGBoost agrees: 62% of its SHAP attribution is the officer record.

One decision, rule by rule: the two highest-scored test searches (log-odds)

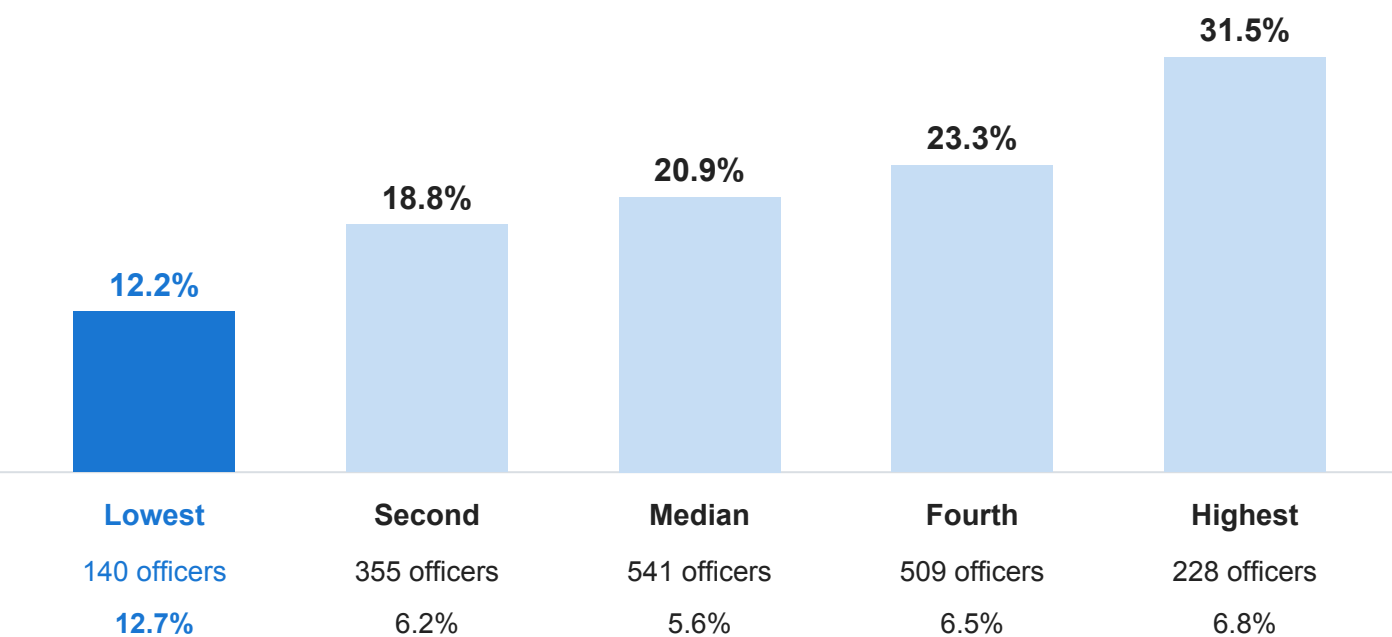
	Found contraband	Found nothing
Predicted probability	0.76	0.75
Driver	white woman	white man
Officer's past consent hit rate	+2.37	+2.36
Experienced officer, first stop of the day	+0.21	+0.21
Driver is white	+0.19	+0.19
Officer's past search rate	+0.06	+0.06
Other rules	+0.04	+0.02
Baseline	-1.73	-1.73
Total log-odds	+1.14	+1.11

Near-identical officer profile, near-identical score, opposite outcomes. In both, the officer's past consent hit rate is above 50% (16.1% across the force in training): the score describes who searches.

PLTR (Dumitrescu, Hué, Hurlin & Tokpavi 2022): a logistic regression on data-driven threshold rules, adaptive lasso, 1-SE rule. The score is exact: log-odds = baseline + the rules that apply. Linear terms: odds ratio per standard deviation.

A fifth of consent searches, made by 140 officers, find contraband 12% of the time and target Hispanic drivers twice as often

Hit rate of 2016–2018 consent searches, by fifth of the officer's past consent hit rate



Share of searched drivers who are Hispanic, by fifth (row above)

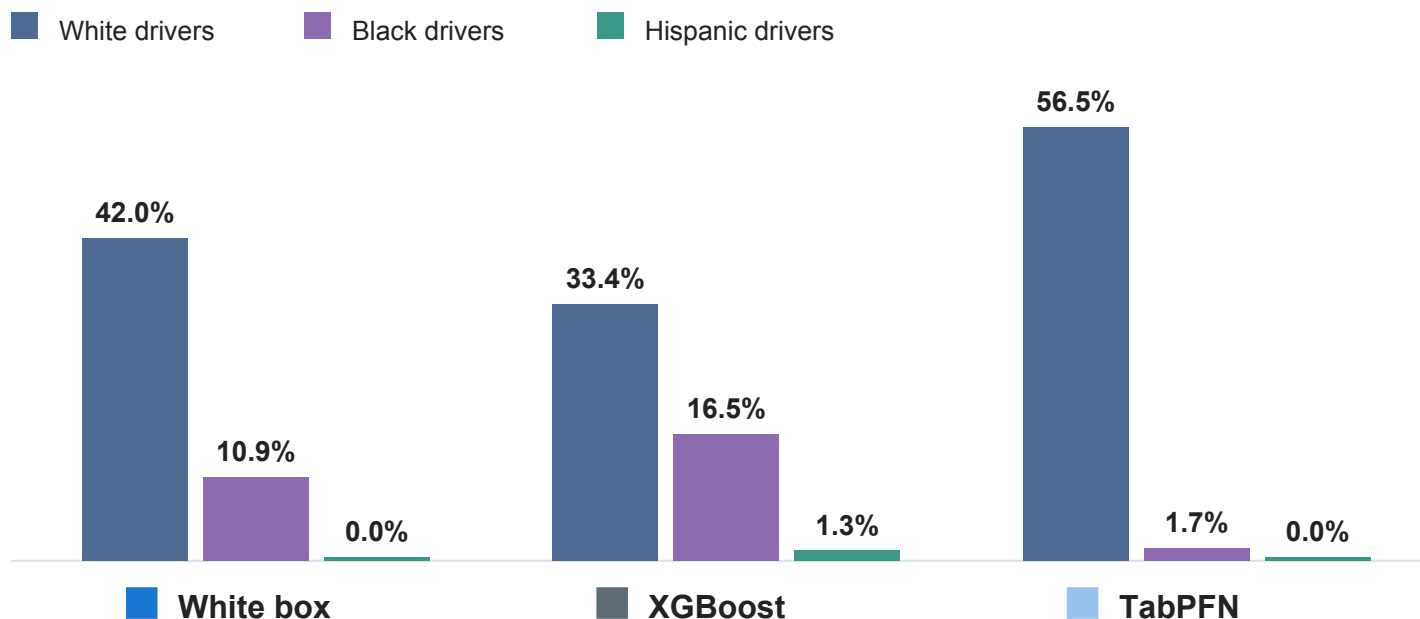
Each officer compared with themselves
1.4× and 2.3×: the median officer asks Black and Hispanic drivers for consent that much more often per stop than white drivers.
Skewed officers find less, on white drivers too: most vs least skewed fifth hit 14.3% vs 21.7% (Black/white skew), 5.0% vs 21.8% (Hispanic/white).
9 officers carry out a quarter of all consent searches of Hispanic drivers (1,146 of 4,645); 98% of those find nothing.
77% of officers find less on Black than on white drivers; 91% less on Hispanic drivers.
Consistent with a lower evidentiary bar. The outcome test cannot prove intent.

Test searches 2016–2018 in fifths (≈ 1,823 each) of the officer's past consent hit rate, known before the search. Skew: each officer's own consent-search rate by driver race, per stop.

Fairness

No version of the model treats every group fairly, because the searches it learned from didn't either

Share of each driver group flagged, at the search budget used earlier



Why: white drivers' searches found something more often in these years (25.5% vs 19.5% Black, 15.1% Hispanic) — a model chasing accuracy copies that.

Hiding race from the model only trades one unfairness for another

Blind it to race, and it flags each group at close to the same rate.

But then the Black and Hispanic drivers it flags are 5–7 points less likely to actually be carrying anything than the white drivers it flags.

In plain terms: it can be fair on paper, or accurate — not both, because the groups' real rates already differ.

The fix is not a better model. It is the search practice behind the data — which is exactly what we recommend acting on next.

"Race-blind" removes the driver's race, sex and same-race feature; the officer's own past search-skew features remain (a fully blind rerun is pending). XGBoost shows the same pattern (MASTER §6.2). Full numbers, with confidence intervals, in backup.

Accuracy barely separates the three models; explainability, stability and governance do

Dimension	White box	XGBoost	TabPFN
Accuracy	Slightly behind, barely above random either way	Statistically tied with TabPFN	Statistically tied with XGBoost
Explainability	Exact — the model is its own explanation	Exact, but a simple version only captures 39% of it	Approximate only; no real explanation
Stability on retrain	Most stable core	Stable, but its reasons shift over time	Most sensitive of the three
Fairness	Fails: too few minority drivers flagged, or wrong about them if blinded	Fails the same way	Fails worst: 56.5% of white drivers flagged vs 1.7% of Black drivers
Governance	Runs anywhere, easy to audit	Needs standard ML infrastructure	Needs a GPU, or sends data to a third party

The white box wins on everything except a small amount of accuracy — and that small gap buys nothing once we get to how it would actually be used.

MNPD should use the white box to review consent-search practice, not to score drivers at the roadside

Where a model can legitimately help — and where it can't

Use		Why
Tell an officer whom to search	✗	At chance inside one officer's stops; otherwise mostly race
Justify a search after the fact	✗	Would launder a discretionary decision
Decide where to patrol	✗	Barely predicts anything; risks a feedback loop
Review search practice, by officer	✓	This is our recommendation
Train officers on real patterns	✓	Patterns are readable and repeat
Report to oversight, year on year	✓	Tracks the disparity honestly

What the review looks like, per officer, each quarter

- 1 How often they ask for consent to search
- 2 How often that search finds anything
- 3 Whether that differs by the driver's race
- 4 How they compare with similar officers



Why the white box, not a black box
Reviewing someone's conduct has to be explainable and contestable
The black boxes' small accuracy edge buys nothing here
Runs anywhere; no data leaves MNPD

Illustrative counterfactual: the least successful fifth keeps its finds and drops searches until it reaches the median fifth's hit rate; 95% CI by resampling officers. "About two thirds" assumes the avoided searches are spread like that fifth's searches.

We ask MNPD to approve a six-month pilot of the officer-level review

Our answers to the five questions

#	Question	Answer
1	Can a model predict a successful search?	No: AUC 0.53–0.55, near random economically
2	What do the models learn?	Mostly race; only the white box is exact
3	Are they stable and fair?	No: unstable shortlists, no version is fair
4	Where do outcomes come from?	The searching officer's practice
5	Which model, for what?	The white box, for officer-level review; no roadside score

What we’re asking MNPD to approve

A six-month pilot

Human review only — no automatic sanctions, and officers can see and contest their report
Retrained every year, and monitored for drift
Method published to the oversight board, plus the threshold test as the next check

This also satisfies the trustworthy-AI requirements the course asks for (human oversight, transparency, fairness) that a roadside score fails — full checklist in backup.

Issue whether to launch the supervisor-review pilot and publish the method to the oversight board

Limits: selective labels (only searches made are observed); the outcome test cannot prove intent (infra-marginality); blank contraband fields were recorded as "nothing found".

Only the officer record adds signal, and it uses strictly past events

Block	Content	Predicts contraband?	Note
Stop and driver	Age, sex, race, precinct, zone, reason for stop, plate state, hour, weekday, month	AUC 0.55–0.57	The task as posed (Act I)
Time	7 cyclical and calendar features	+0.007	The only block that helps in Act I
Neighbourhood (ACS)	800 m / 1,200 m population-weighted circles, release before the stop year	Adds nothing once fixed	As first shipped, a year fingerprint (AUC 0.998 for the 2016 split); predicts race at 0.69–0.73
Plate	State, region, distance	No	Re-encodes plate state; plate_missing is a 2017 recording change
Officer record	25 features from the searching officer's own earlier stops	+0.07 pooled AUC	A measuring instrument, not a deployable input

Leakage controls

The 11 post-decision columns carry a post_ prefix and are blocked.
Officer features use only events strictly before the stop: same-minute stops never see each other, the stop's own outcome is never counted; checked by brute force on 400 random stops.
PLTR re-learns its rules inside each cross-validation fold; on pure noise it gives CV AUC 0.505.

The officer-record result holds across years, unseen officers and resamples

Check	White box (PLTR, 19 rules)	XGBoost + officer record
Test AUC 2016 / 2017 / 2018	0.625 / 0.651 / 0.605	0.637 / 0.648 / 0.628
Officers never seen in training	0.629	0.643
5 other draws of held-out officers	0.618 ± 0.026	0.635 (0.595–0.695)
Resampling	50 bootstraps: AUC 0.630 ± 0.004; 0 sign changes among selected rules	10 seeds 0.640 ± 0.002; 20 bootstraps 0.635 [0.628, 0.641]
What stays	Officer consent hit rate selected in every resample (OR 1.25–1.48); rules sharing variables carry 62% of importance	Importance ranking correlates 0.94 across bootstraps, 0.85 across eras
Drift	AUC drops in 2018	Top officer features drift strongly (PSI 0.75–0.95)

Consequence for deployment: retrain yearly and monitor drift (slide 13).

The white box in full: 19 rules, read as a sum of log-odds

Rule	Odds ratio	Share of searches	Share of importance
Officer's past-year consent hit rate (per SD)	1.43	100%	32.0%
Officer's past hit rate low on this driver's race and overall	0.64	15%	14.4%
Driver is Hispanic	0.71	8%	8.3%
Driver is white	1.21	36%	8.2%
Experienced officer, first stop of the calendar day	1.24	21%	7.8%
Officer's previous search < ~110 min ago	0.79	10%	6.3%
Male driver, officer with ≤ 9 consent searches last year	0.85	19%	5.5%
Lower-volume officer, stop at ~4:30–7:30	1.46	2%	5.3%
Officer searches Hispanic drivers > 1.6× as often as white	0.93	21%	2.7%
Low search-rate officer, driver aged 19 or younger	1.09	9%	2.3%
Lower-volume officer, stop after 21:30	1.06	14%	1.9%
Lower-volume officer, September to December	1.04	29%	1.5%
Zone 125	1.12	1%	1.0%
Officer's past-year search rate (per SD)	1.01	100%	0.8%
Seven precincts, stop at ~23:00–13:00	0.98	27%	0.7%
Stopped for a moving traffic violation	1.01	39%	0.6%
Lower-volume officer, late in the week	0.99	32%	0.5%
Routine violation, stop outside ~10:00–14:00	1.00	42%	0.1%
Zone 113	1.01	1%	0.1%

Searches at the officer's first stop of the calendar day hit more often, in every time band

Hit rate of consent searches, first stop of the day vs later stops, by hour band



33%

of consent searches happen at the officer's first stop of the calendar day

Hit rate 20.9% at the first stop vs 15.8% later

34% of first-stop searches fall between 00:00 and 03:59: this is not "start of shift"

Holds for low-, mid- and high-volume officers: 21% vs 16%, 22% vs 16%, 19% vs 15%

First = no earlier stop by this officer on the same calendar day. Reading: later, repeated searches look more speculative; a hypothesis for training, not a causal claim.

Act I in detail: every model flags mostly white drivers, and TabPFN is the most skewed

Model	Drivers	Flagged %	False-positive rate %	Hit rate of flagged %	Predicted – observed pp
White box	White	42.0 [40.4, 43.8]	42.6 [40.9, 44.5]	24.4 [22.2, 26.6]	–4.1 [–5.6, –2.7]
	Black	10.9 [10.0, 11.7]	10.9 [10.0, 12.0]	19.0 [15.7, 22.2]	–3.3 [–4.4, –2.1]
	Hispanic	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	n/a	–10.7 [–13.4, –8.2]
XGBoost	White	33.4 [31.9, 35.0]	32.2 [30.3, 34.1]	28.3 [25.9, 31.0]	–3.9 [–5.5, –2.3]
	Black	16.5 [15.5, 17.5]	16.3 [15.3, 17.4]	20.2 [17.6, 22.9]	–4.7 [–5.7, –3.5]
	Hispanic	1.3 [0.6, 2.2]	1.0 [0.3, 1.9]	33.3 [0.0, 71.4]	–11.3 [–14.2, –8.7]
TabPFN	White	56.5 [54.9, 58.1]	56.4 [54.4, 58.2]	25.7 [23.8, 27.8]	–4.9 [–6.4, –3.6]
	Black	1.7 [1.3, 2.0]	1.7 [1.3, 2.1]	18.3 [9.8, 27.5]	–2.6 [–3.6, –1.6]
	Hispanic	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	n/a	–10.0 [–12.5, –7.6]

χ^2 (selection, Black vs white): white box 1,081 · XGBoost 322 · TabPFN 3,314

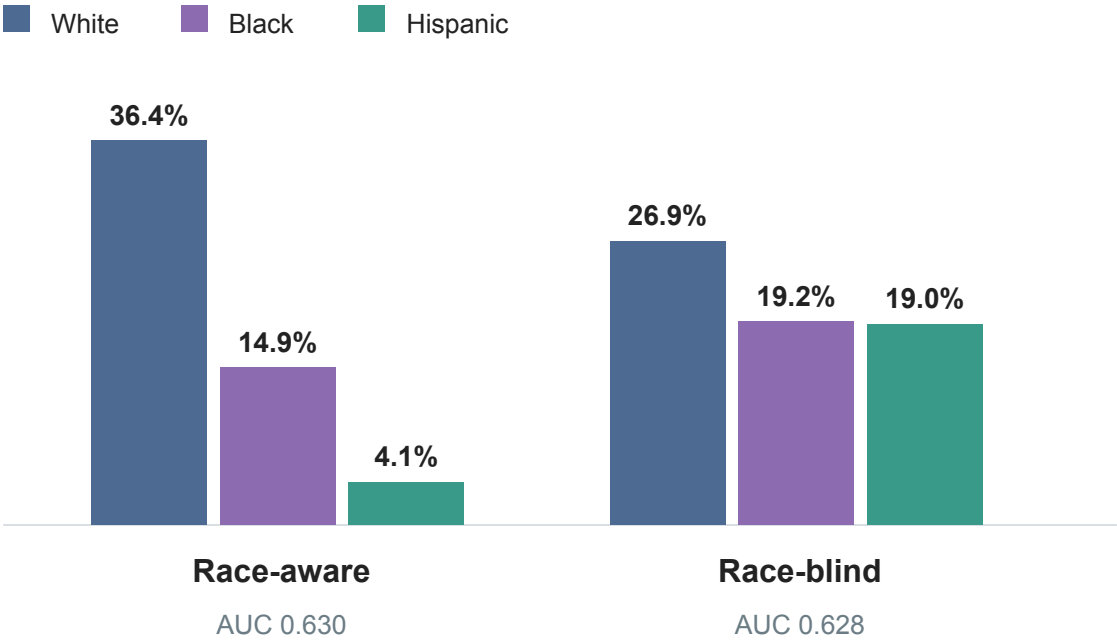
Hit-rate gap at equal score, Black – white (pp): white box –6.0 · XGBoost –4.6 · TabPFN –5.9

Predicted – observed is negative everywhere: models trained on 2010–2015 (16.1% hit) under-predict 2016–2018 (21.4%); compare groups, not levels.

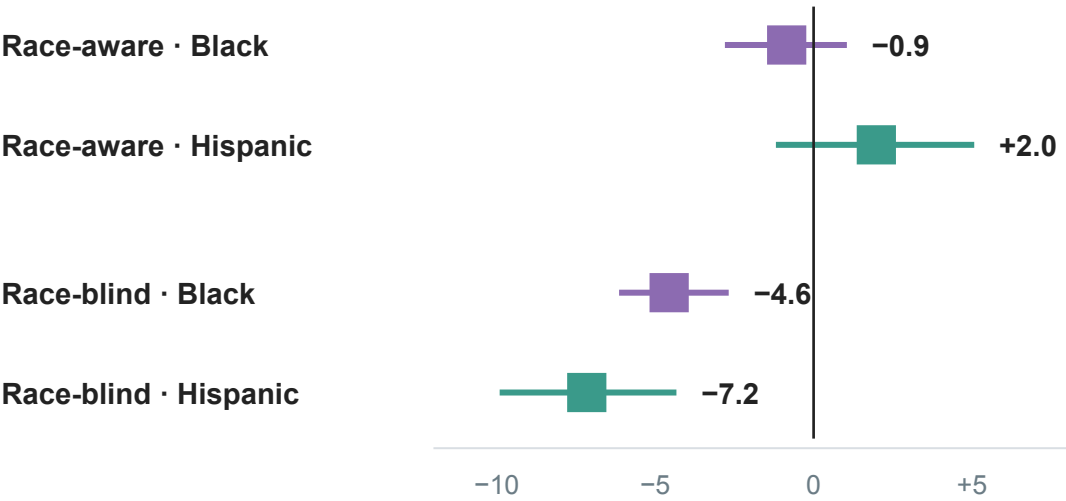
Backup

Removing race: selection evens out, but the model becomes wrong about minority drivers

Share of each group flagged at the top 22%, white box with officer record



Hit-rate gap vs white drivers at equal score (pp, 95% CI)



≈ 0: calibrated. Below 0: at the same score, minority drivers carry contraband less often, so they are over-scored.

Test hit rates differ: white 25.5%, Black 19.5%, Hispanic 15.1%. No single feature restores parity (fairness partial dependence: 0 of 32). The disparity is in the labels, produced by search decisions, so the fix is upstream, in search practice.

The trustworthy-AI checklist, in full (EU High-Level Expert Group, 2019)

Requirement (EU HLEG 2019)	Roadside search score	Officer-level review (white box)
Human agency and oversight	✗ Pre-empts a discretionary legal decision	✓ A supervisor decides; the model only flags
Technical robustness	✗ Near-chance, unstable shortlist	✓ Officer records persist, transfer and hold every year; retrained yearly for drift
Privacy and data governance	✗ Scores every driver	✓ Aggregates per officer; hashed IDs in the analysis; no third-party API
Transparency	✗ Race drives the signal; TabPFN cannot be explained	✓ 19 readable rules; published method
Non-discrimination and fairness	✗ Reproduces or inverts the disparity	✓ Targets the practice that produces it
Societal well-being	✗ More searches of the wrong people	✓ Fewer fruitless searches, most of them of minority drivers
Accountability	✗ Who answers for a score?	✓ Review protocol, appeal for officers, oversight reporting

Officers are subjects too: intervals and shrinkage on every rate, false-discovery control across ~1,500 officers, human review only, the right to see and contest one's report, no automatic sanctions.

TabPFN: what we can claim, and what we cannot

Reproducible from src (Act I)

AUC 0.553 matched, 0.573–0.580 with all training rows
Brier 0.169, still worse than a constant (0.168)
Within officer 0.533 [0.512, 0.553]
Permutation importance: race first (0.038)
Flags 56.5% / 1.7% / 0% of white / Black / Hispanic drivers (χ^2 3,314)
Resampling its training context moves AUC by 0.024

Run interactively (Act II), reported as is

AUC 0.653 with all 25 officer features (XGBoost on the same 25: 0.646, vs 0.640 with 16)
0.650 without driver race but with 8 race-derived officer features: not race-blind
Within-officer AUC, economics and fairness not computed (no predictions saved)

Governance

209 s on CPU to score 9,113 searches
The hosted API sends stop records to a third party
No native attribution; occlusion error cannot be bounded
No global surrogate, by choice: on XGBoost a depth-3 tree reproduces only 39% of the score (course slide 86: "illusion of interpretability")

Glossary

Term	Meaning
AUC	Probability that the model ranks a random successful search above a random fruitless one; 0.5 = random
Within-officer AUC	AUC inside each officer's own searches, averaged (weighted by searches): the value of a score for the decision an officer actually faces
Break-even cost ratio	Hit rate / (1 – hit rate) of the searches kept: above this ratio of (cost of a fruitless search / value of a find), searching destroys value
Outcome test	Compare hit rates by group: a lower hit rate is consistent with a lower bar for searching that group
Infra-marginality	Why the outcome test cannot prove intent when groups' risk distributions differ; the threshold test addresses it
Selective labels	The outcome is observed only for the searches officers chose to make
Calibration at equal score	Among drivers with the same score, do groups carry contraband equally often?
XPER	Shapley decomposition of the AUC above 0.5 across features (course method)
PLTR	Penalised logistic tree regression: a logistic regression on data-driven threshold rules
χ^2 (selection)	Test that the share flagged is independent of the driver's race